

		$0.060(v + 45) = 95 \times 70 \times 10^{-3}$ $v = 66 \text{ ms}^{-1}$
7	C	Regardless of the nature of the wall, it will exert a normal contact force (horizontally) on to the ladder. To balance it, a horizontal force acting to the left must be provided by the ground. That is only possible if the ground is rough so as to provide a frictional force on the ladder.
8	B	$U_w + U_m = W$ $\rho V_1 g + \rho V_2 g = \rho V_3 g = \rho (V_1 + V_2) g$